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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-54

DC Charging (CCS2)-V

Dr. Apurv Kumar Yadav

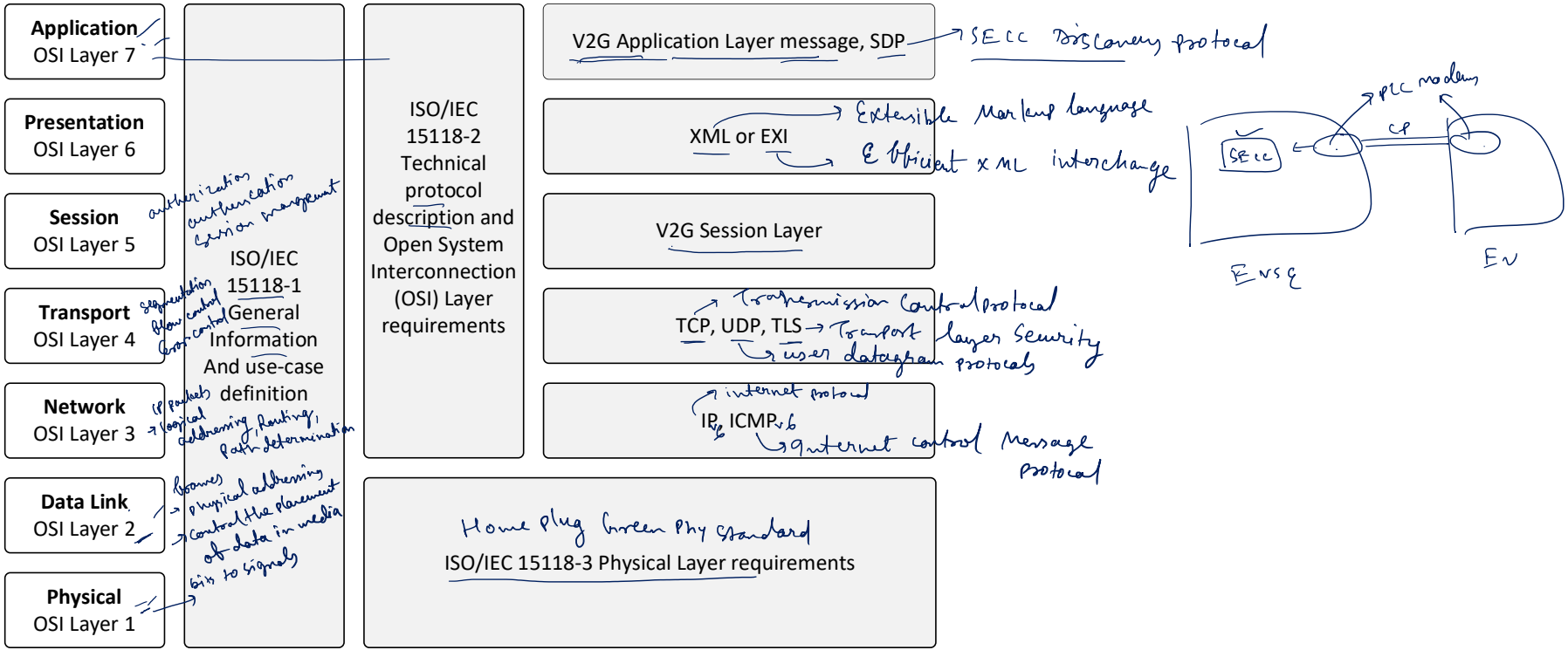
Department of Electrical Engineering



Recap

- The 7 OSI layer Model
- HLC with CCS2

Recap: OSI Layer Model in CCS2



Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer1 (Physical)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
• PE & Control Pilot connected • PWM of 3-7%; State: "B2" • On both sides: compatible Home Plug Green PHY modems	Establish physical link on the PE-CP wires to the opposite side	PLC module ready for communication with an established frequency band of 2 MHz to 28 MHz

Layer ensures the start of physical connections to provide data transfer between EV and EVSE, also defines physical properties of the physical media such as voltage levels, frequency, etc.

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer2 (Datalink)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established Physical layer	• Configuration of PLC nodes • Signal Level Attenuation Characterization (SLAC) procedure • Set-up logical connection of PLC Modems	Establishment of communication Link

Layer ensures an error-free transfer of data frames from one node to another over the physical media

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer2 (Network)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established datalink layer	• Implement the IPv6 Internet Protocol Standard to define IP address • Ensure unique addresses • Implementation of ICMPv6 to send error messages and SLAAC procedure	All nodes obtains the valid IP address by using the SLAAC procedure (Stateless Address Auto configuration)

Layer controls the routing of connections which decides the path for the data which it should take

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer2 (Transport)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established network layer	• All the nodes implement Transmission Control Protocol (TCP) for reliable, User Datagram Protocol (UDP) for fast, Transport Layer Security (TLS) (optional) for safer	Establishing reliable, fast and safe data connection between nodes

Layer ensures fast, reliable and safer communication between the nodes

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer2 (Session)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established transport layer	• Nodes implement V2G transport Protocol (V2GTP) • Establishes, manages and terminates connections between the nodes by using IP addresses and port numbers • Implement session layer header which contains some basic information like protocol version, payload type, payload size	Establishment of charging session connections between nodes ↓ EV & EVSE

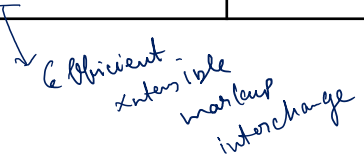
Layer ensures establishing the session for data exchange between nodes (EV & EVSE)

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Recap: CCS2- HLC: OSI Layer2 (Presentation)

- Included in ISO 15118 standard

Initial Condition	Function	Outcome
Successfully established session layer and both EV and EVSE use encoding format using EXI	• Coding and decoding from application to network format • The V2G EXI protocol consists of two different message sets: V2G application layer protocol handshake messages, V2G application layer messages • Implement message header (contains Session ID, Notification, Signature) and message body (represents the abstract message information)	validity and compatibility of exchanged messages is ensured



Layer transforms data which enables correct data exchange between EV and EVSE. Act as the translator of the system

Source:
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Recap: CCS2- HLC: OSI Layer2 (Application)

- Included in ISO 15118 standard

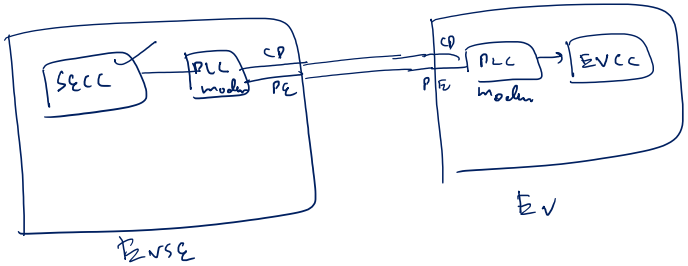
Initial Condition	Function	Outcome
Successfully established presentation layer	• Manages the messages between EV and EVSE • Establish charging process (i.e. identification, pre-charge, charge, security check, etc) • The application protocol messages used to select the charging standard between • EV sends the **supportedAppProtocolReq** (contains all standards which EV supports). • The EVSE selects one of the standards by sending the **supportedAppProtocolRes** with the selected SchemeID	Representing the messages and the required communication protocol _

Layer ensures the initialization and configuring of the charge process.

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

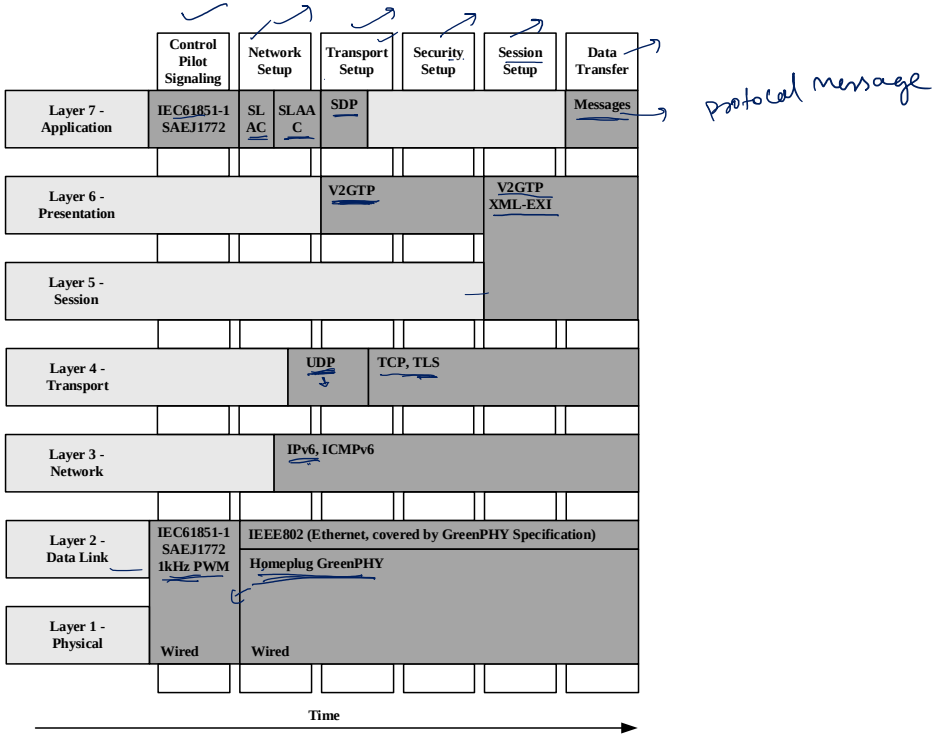
CCS2- HLC: SECC Discovery Protocol

- After the physical connection (plugged in) and the logical connection (SLAC procedure) is established, the EV has to find the supply equipment communication controller (SECC).
- The SECC may be placed in a data center of the charging station service provider.
- To get the IP address of the SECC, the EV uses the SECC Discovery Protocol (SDP) → SDP is a UDP based protocol
- The SDP Request is sent via broadcast (from EV) to the SECC which contains the session ID.
- The SDP Response message (from SECC) contains the SECC IP, session ID and SECC TCP port which is used for the communication.



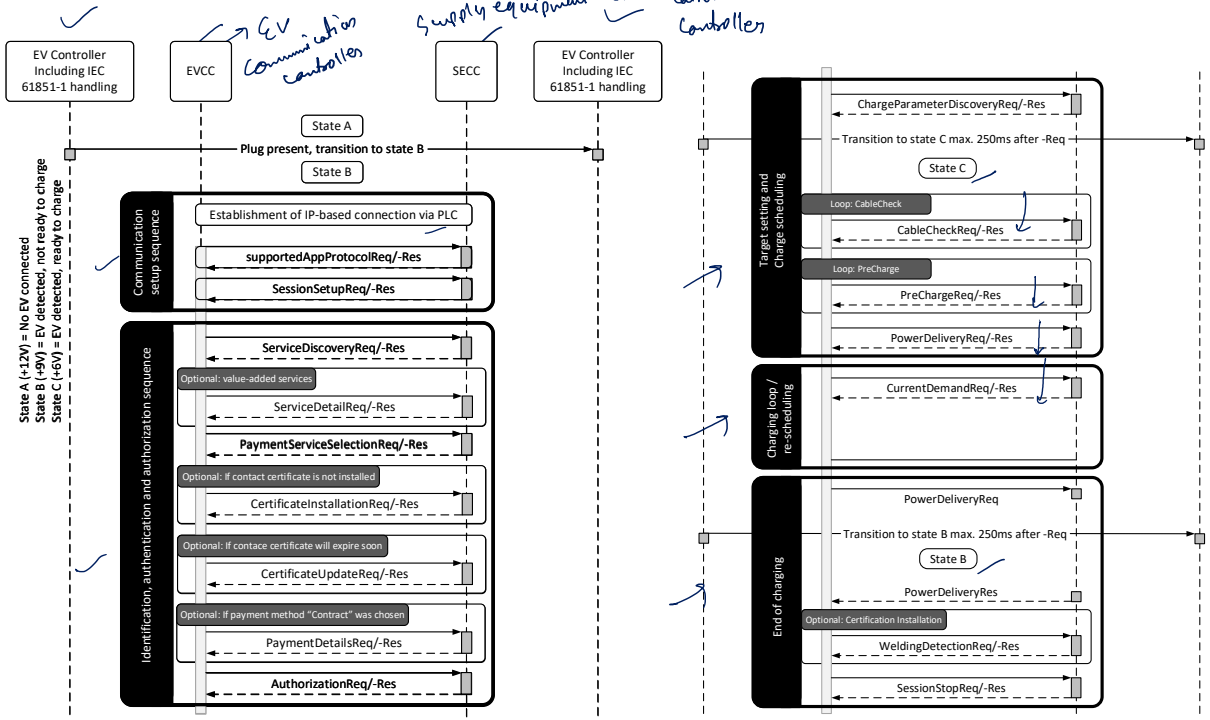
Courtesy:
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CCS2- HLC



Courtesy: Keysight;
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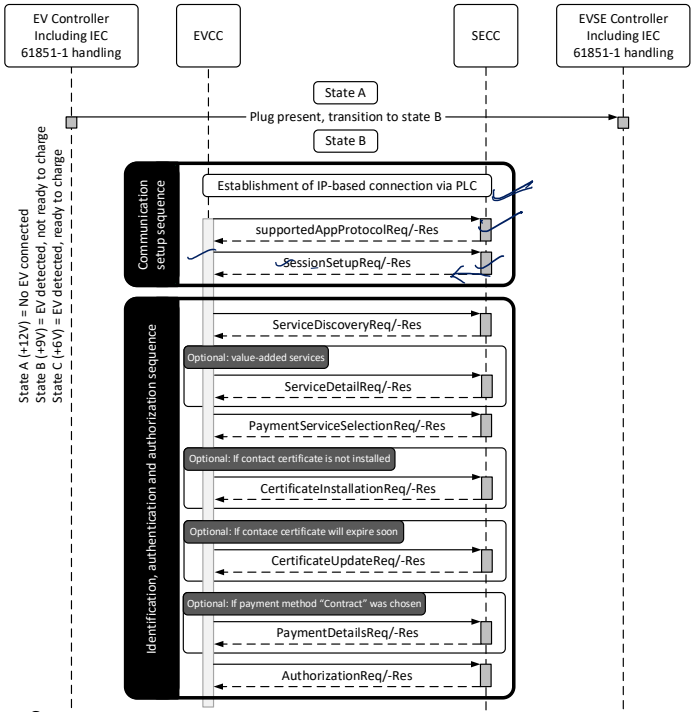
CCS2- HLC: Protocol Messages (ISO 15118/DIN SPEC 70121)



- Simple request-response protocol
- EV is always the master (request)
- EVSE is the slave (response)
- In case of critical error, EVSE will answer wrong requests with Response code = FAILED Sequence Error
- After an error, the session shall be closed immediately

Source:
https://www.typhoon-hil.com/documentation/typhoon-hil-software-manual/References/iso15118_protocol.html

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)



Session Setup

- Create new session
- EV sends session setup request with session ID as 0
- EVSE sends session setup response with new session ID

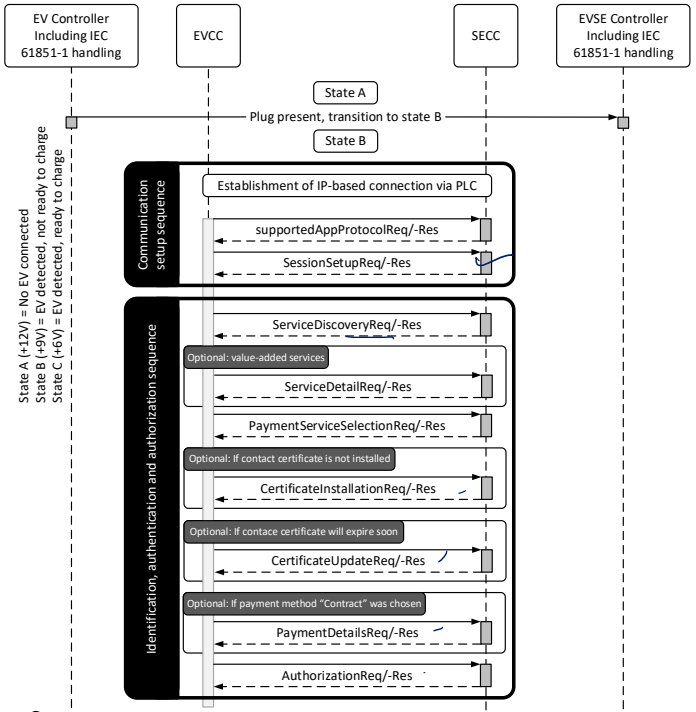
⇒ session ID will be then used during the entire communication process

⇒ If EV sends invalid session ID, EVSE sends the response with Response Code failed (SessionSetup response message)

⇒ Restart of session (only in ISO15118)
EV send the session setup request with the last session ID which was used & EVSE will then respond with the same session ID

Source:
https://www.typhoon-hil.com/documentation/typhoon-hil-software-manual/References/iso15118_protocol.html

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)

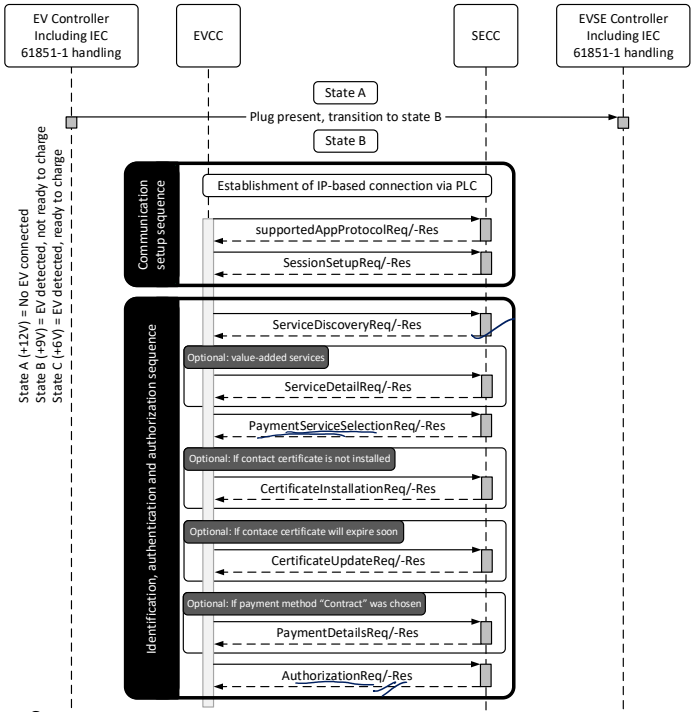


Source:
https://www.typhoon-hil.com/documentation/typhoon-hil-software-manual/References/iso15118_protocol.html

Service Discovery

- EV request to EVSE about the information of available services.
- For example, a basic service is "EV Charging", "payment" .
- ISO 15118: All details about services are requested with **ServiceDetailsReq/Res** messages
- Service details contains all the information regarding the service

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)



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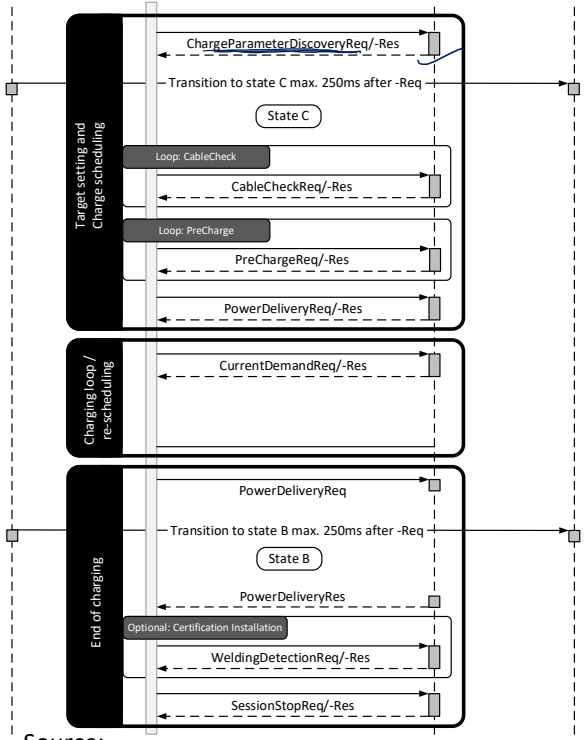
ServicePaymentSelection (DIN SPEC 70121) or PaymentServiceSelection (ISO 15118)

- EV is requesting the type of payment (Contract or External)
- EV adds a list of services which should be used

Authorization:

- Messages include: ContractAuthentication (DIN SPEC 70121)/ Authorization (ISO 15118)
- Waiting until the session is authorized by EVSE.

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)



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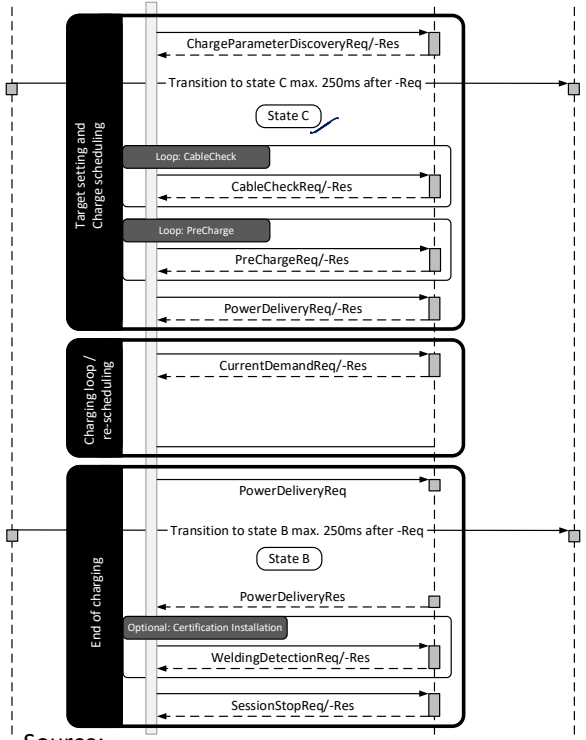
Charge Parameter Discovery

- Exchange of charging parameters such as maximum voltage, current, power
- EV sends the charging parameter discovery request message multiple times until the EVSE sends a response.

↳ EVSE Processing = Finished

EVSE Processing = "ongoing" ; if parameters are not determined

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)



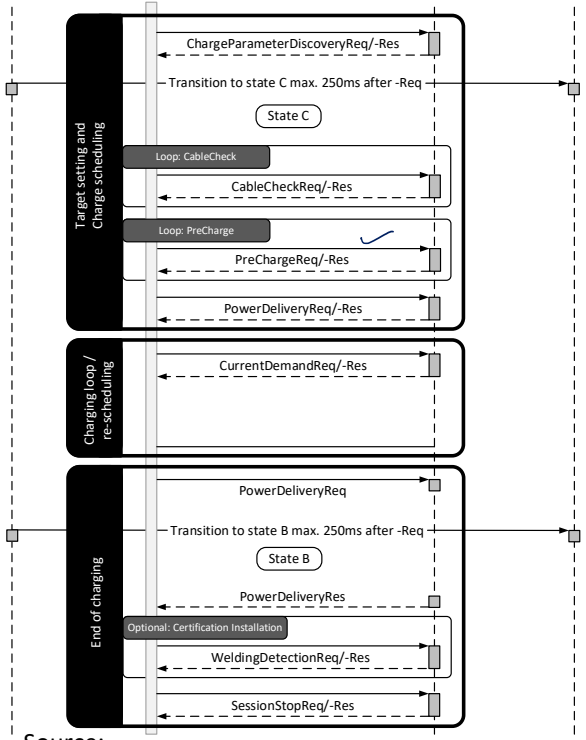
Source:
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Cable Check

- The Cable Check is done with continuous monitoring of the insulation resistance.
- The cable check request is sent by the EV multiple times during EVSE operations.
- The EVSE sends cable check response.

EVSE processing = finished (if completed)
= ongoing (if processing is not completed)

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)

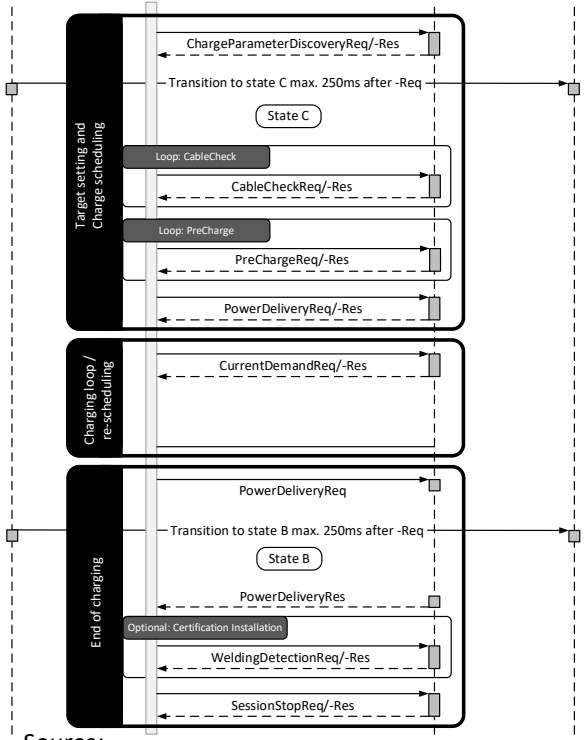


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PreCharge

- The Pre charge request is sent by EV until the EV can measure a voltage on EVSE side which matches the battery voltage (with a tolerance = 20V).

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)

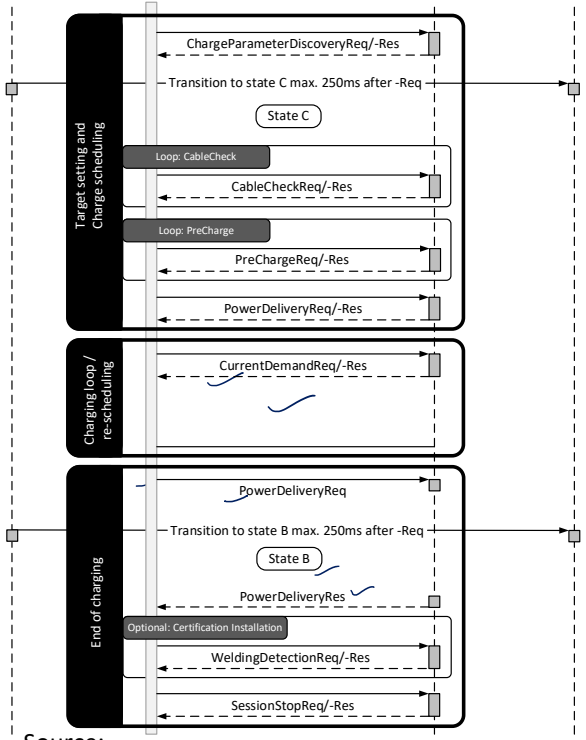


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Power Delivery

- The Power delivery request is sent by the EV after the Pre Charge is successfully done exchanged.
- After receiving the Power delivery response , the EV closes its contactor switch and proceeds with the charging process.

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)

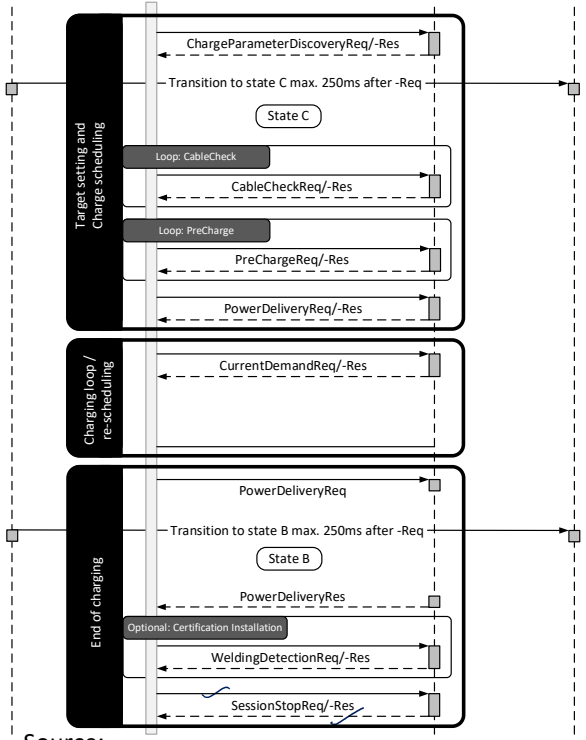


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Current Demand

- The Current demand request is sent by EV continuously during the charging process
- Each request contains a request for the charging voltage and the charging current
- EVSE processes the demanded values and set the power source to the required values.
- EVSE can trigger the stop of charging by sending a notification inside of the Current demand response
- EV stops the charging process

CCS2- HLC: Protocol (ISO 15118/DIN SPEC 70121)



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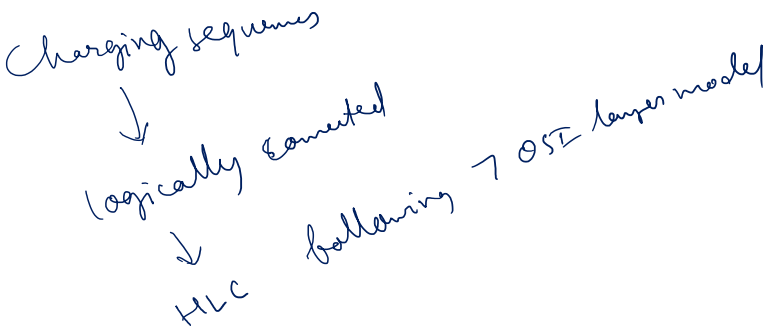
Welding Detection

- Welding Detection is an optional step
- Welding detection request is sent by EV
- Welding detection response is sent by EVSE which contains the measured voltage.

Session Stop

- To stop the session

CCS2 DC Charging System



Thank You